

☒ Easy Level (10 Questions)

Q1. How do you create an empty dictionary?

- A. []
- B. {}
- C. ()
- D. dict[]

Answer: B

Explanation: {} creates an empty dictionary.

Q2. Which method adds a new key-value pair or updates an existing key?

- A. add()
- B. update()
- C. append()
- D. insert()

Answer: B

Explanation: update() adds or modifies dictionary items.

Q3. What keyword deletes a key from a dictionary?

- A. delete
- B. remove
- C. del

D. pop

Answer: C

Explanation: del dict[key] removes a key.

Q4. Which mode is used to open a file for writing?

A. 'r'

B. 'w'

C. 'a'

D. 'x'

Answer: B

Explanation: 'w' opens file for writing (creates or truncates).

Q5. Which data type uses square brackets?

A. List

B. Tuple

C. Dictionary

D. Set

Answer: A

Explanation: Lists use [].

Q6. How do you add an element at the end of a list?

A. append()

- B. insert()
- C. add()
- D. extend()

Answer: A

Explanation: append() adds single element at end.

Q7. What does pop() do when used on a list?

- A. Deletes entire list
- B. Removes and returns last element
- C. Deletes a specific element by value
- D. Finds element position

Answer: B

Explanation: pop() removes and returns last item by default.

Q8. Which operator concatenates two lists?

- A. *
- B. +
- C. -
- D. /

Answer: B

Explanation: + joins two lists.

Q9. How do you check if a key exists in a dictionary?

- A. in operator
- B. has_key() method
- C. exists()
- D. find()

Answer: A

Explanation: Use key in dict.

Q10. Which of the following is immutable?

- A. List
- B. Dictionary
- C. Tuple
- D. Set

Answer: C

Explanation: Tuples are immutable.

☒ Medium Level (20 Questions)

Q11. What happens if you try to access a non-existent key in a dictionary using dict[key]?

- A. Returns None
- B. Returns empty string
- C. Raises KeyError

D. Adds key with None value

Answer: C

Explanation: Accessing missing keys raises KeyError.

Q12. Which method safely retrieves a value from a dictionary with a default if key is missing?

A. get()

B. find()

C. search()

D. lookup()

Answer: A

Explanation: dict.get(key, default) returns value or default.

Q13. How can you open a file for appending?

A. 'r'

B. 'w'

C. 'a'

D. 'x'

Answer: C

Explanation: 'a' mode appends data to end of file.

Q14. What does this code do?

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```
my_list = [1, 2, 3] my_list.insert(1, 10)
```

- A. Inserts 10 at index 1
- B. Replaces element at index 1
- C. Adds 10 at end
- D. Error

Answer: A

Explanation: insert() inserts element at given index.

Q15. What is the output of:

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```
my_dict = {'a': 1, 'b': 2} print(my_dict.get('c', 3))
```

- A. None
- B. 3
- C. KeyError
- D. 2

Answer: B

Explanation: Returns default value 3 since 'c' not present.

Q16. Which of the following will remove the last item from the list and return it?

- A. del list[-1]
- B. pop()
- C. remove()
- D. clear()

Answer: B

Explanation: pop() removes and returns last element.

Q17. How to concatenate two lists a and b?

- A. a.append(b)
- B. a + b
- C. a.extend(b)

D. Both B and C

Answer: D

Explanation: Both `a + b` and `a.extend(b)` combine lists.

Q18. What does this code print?

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```
d = {'x': 1, 'y': 2} del d['x'] print('x' in d)
```

A. True

B. False

C. KeyError

D. None

Answer: B

Explanation: 'x' was deleted; thus not in dict.

Q19. Which of these is NOT a valid file mode?

- A. 'r+'
- B. 'w+'
- C. 'a+'
- D. 'z'

Answer: D

Explanation: 'z' is not a valid mode.

Q20. What is the correct way to read entire content of a file?

- A. f.read()
- B. f.readline()
- C. f.readlines()
- D. f.readall()

Answer: A

Explanation: read() reads full file content.

☒ Hard Level (10 Questions)

Q21. What will this output?

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```
a = [1, 2, 3] b = a b.append(4) print(a)
```

- A. [1, 2, 3]
- B. [1, 2, 3, 4]
- C. Error
- D. [4]

Answer: B

Explanation: a and b refer to same list, so change reflects in both.

Q22. Which of these correctly removes a key from dictionary and returns its value?

- A. dict.remove(key)
- B. dict.pop(key)
- C. del dict.key
- D. dict.delete(key)

Answer: B

Explanation: pop() removes key and returns value.

Q23. What will be the result of this slicing?

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```
lst = [0,1,2,3,4,5] print(lst[2:5])
```

- A. [2,3,4]
- B. [3,4,5]
- C. [2,3,4,5]
- D. [2,3]

Answer: A

Explanation: Slices from index 2 to 4.

Q24. What happens when you try to remove an element not in the list using remove()?

- A. Nothing
- B. Raises ValueError
- C. Removes closest element
- D. Deletes entire list

Answer: B

Explanation: Raises ValueError if element not found.

Q25. How do you safely open a file and ensure it is closed even if an error occurs?

- A. Use try-finally
- B. Use with statement
- C. Manually close file
- D. Use open() alone

Answer: B

Explanation: with automatically closes file.

Q26. Which of the following statements is true about dictionary keys?

- A. Keys can be mutable
- B. Keys must be immutable
- C. Keys can be duplicated
- D. Keys are always strings

Answer: B

Explanation: Keys must be immutable and unique.

Q27. What happens when you assign a new list with * operator?

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```
lst = [0] * 3  
print(lst)
```

- A. [0, 0, 0]
- B. [0, 1, 2]
- C. [3, 3, 3]
- D. Error

Answer: A

Explanation: * repeats list elements.

Q28. Which of the following methods would you use to get all keys in a dictionary?

- A. keys()
- B. items()
- C. values()
- D. get()

Answer: A

Explanation: keys() returns all dictionary keys.

Q29. What is the difference between append() and extend() on lists?

- A. append() adds one element, extend() adds elements from iterable
- B. Both are same
- C. append() adds multiple, extend() adds one
- D. None

Answer: A

Explanation: append() adds single element, extend() adds multiple.

Q30. What does the readline() method do?

- A. Reads entire file
- B. Reads next line from file
- C. Reads all lines into a list
- D. Closes file

Answer: B

Explanation: readline() reads the next line only.